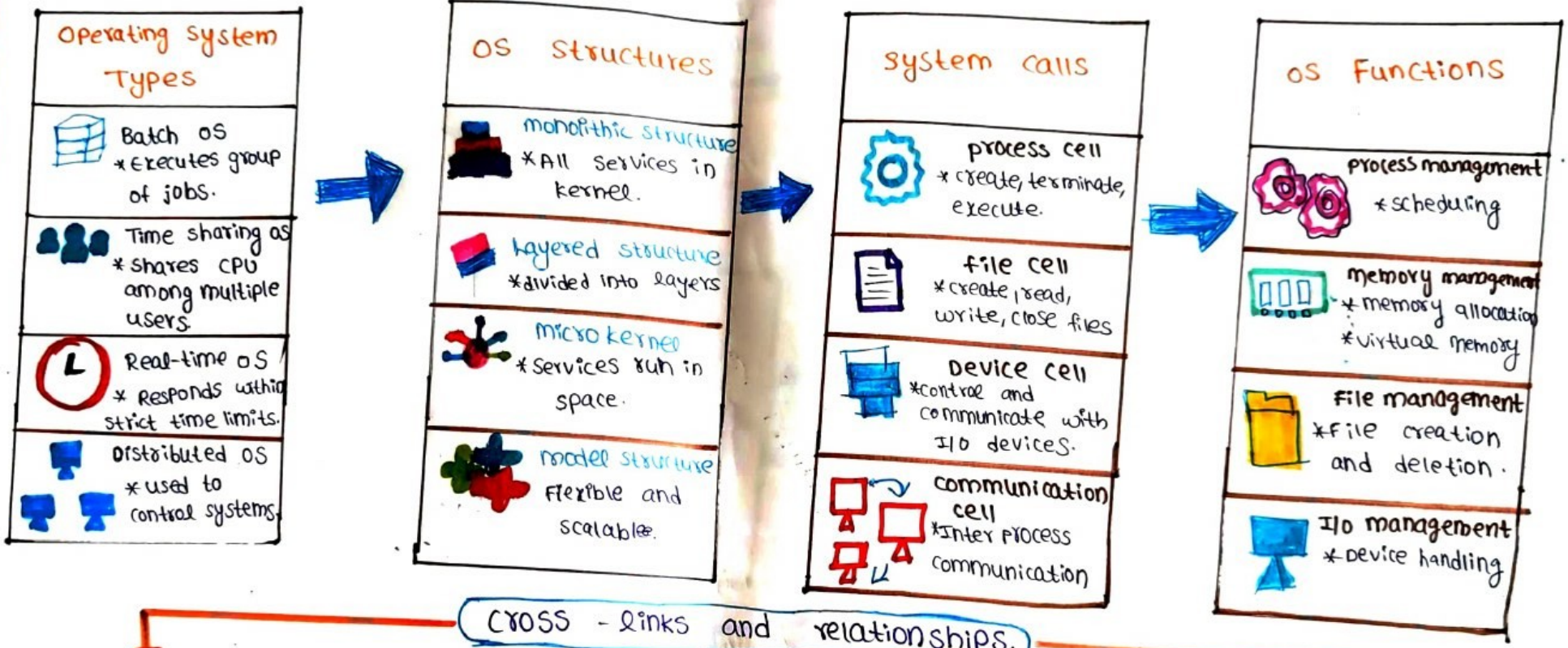
CO1- Assessment tool-2

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1) Operating system Types → OS structures → system calls → OS functions.

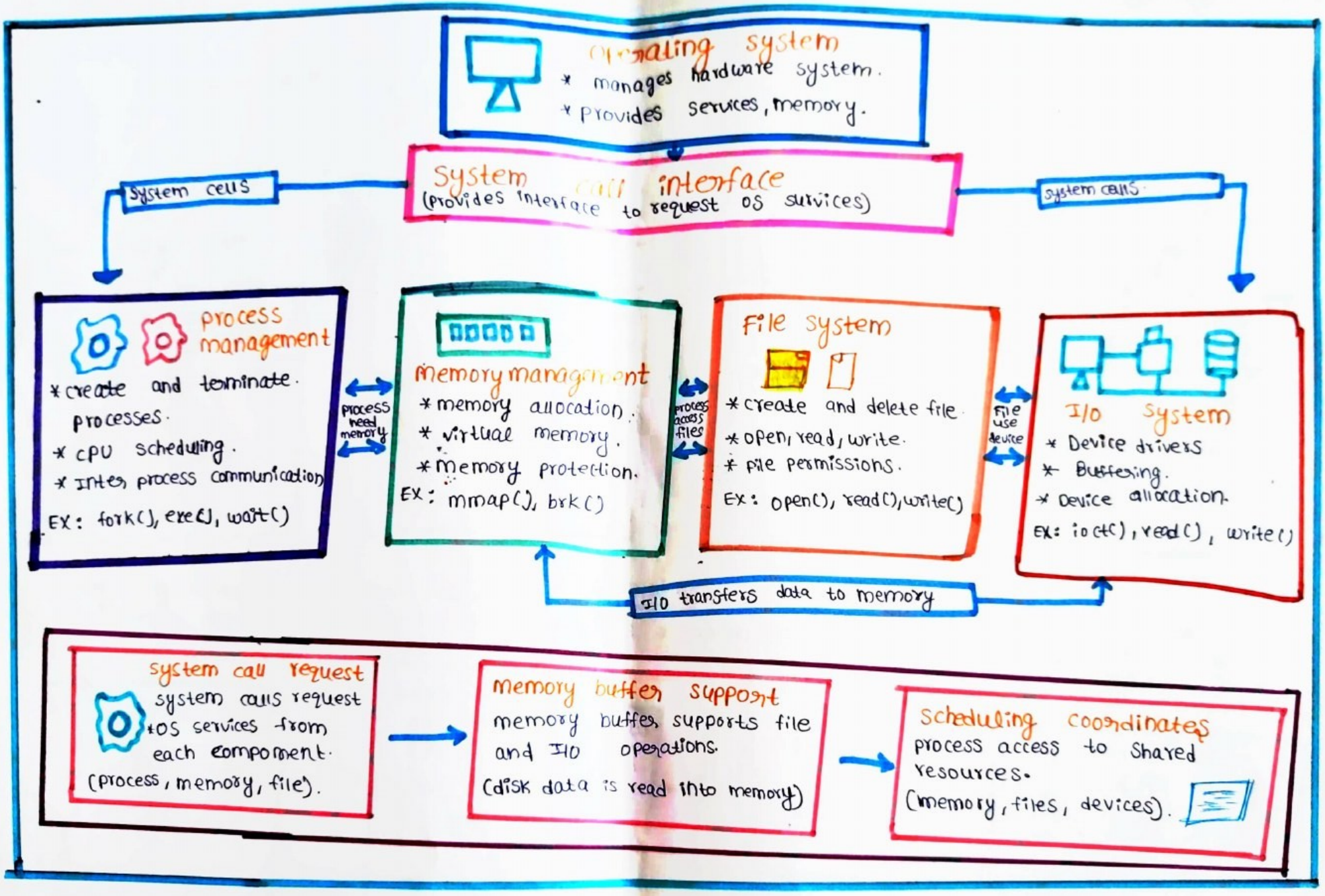


Cross - links and relationships.

OS type decides the Structures and resource management methods → OS structure influences how system calls are implemented. → system calls provides interface to access OS functions. → os functions manage hardware resources and provide services to users.

Comprehensive concept map

2)



Comparative concept map

3)

Operating system Types

* Different OS types are designed for different needs.

Batch OS

* Executes grouped jobs.
* Little / no users.

Advantages

* High job throughput

Limitation

* Slow response time.

Time-sharing OS

* Sharing CPU among users
* Inactive operation.

* **Advantages**
  Quick response time.

Limitation

* Resource contection.

Distributed OS

* Share resource workload
* shares resources.

Advantages

* Resource sharing.

Limitation

* Network dependency.

Real-time OS

* Responds within deadlines.
* used in control system.

Advantages

* predictable timings.

Limitation.

* strict timing.

Similarities.

* All manage harduare resources.
* Execute programs
* manage processes & memory
* provide services to users.

Batch OS

Best for repeated large jobs.

→

Time sharing OS

Best for inactive users.

→

Distributed OS

Best for network. resource sharing.

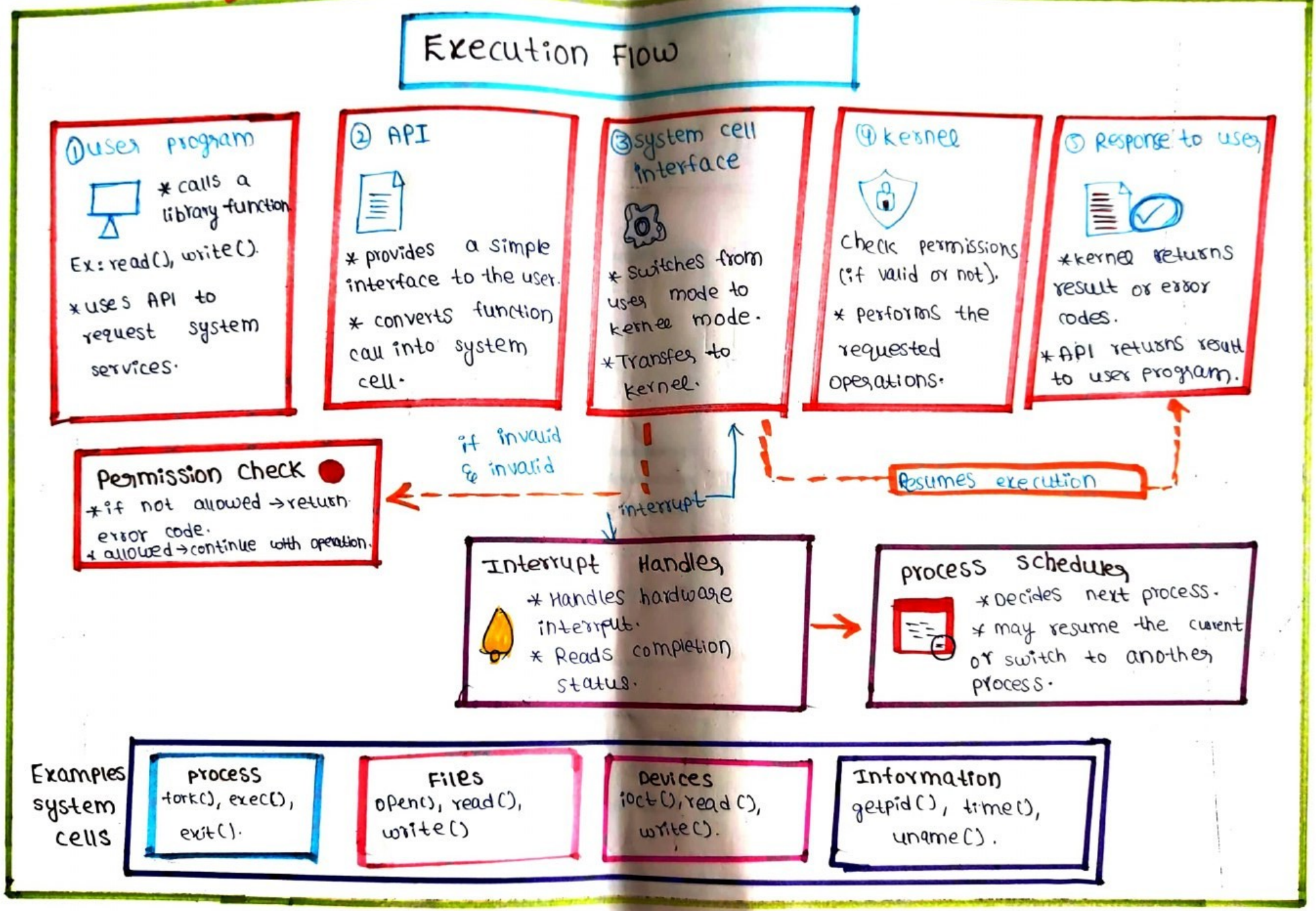
→

Real-time OS.

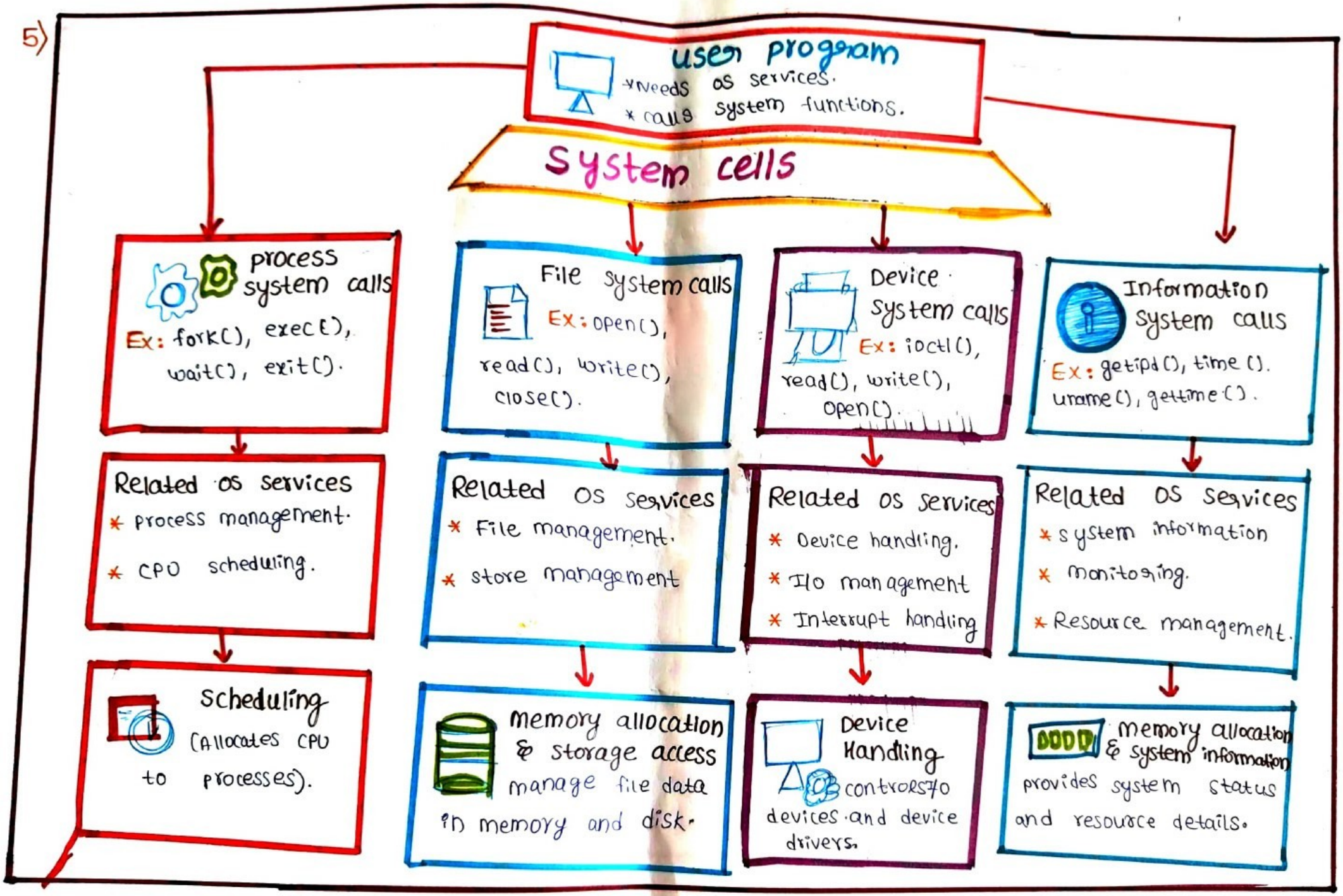
Best for deadline critical tasks.

Dynamic System call Execution Flow

4)



5)


User program
needs OS services.
calls system functions.
System cells
process system calls
Ex: fork(), exec(), wait(), exit().
File system calls
Ex: open(), read(), write(), close().
Device System calls
Ex: ioctl(), read(), write(), open().
Information System calls
Ex: getpid(), time(). uname(), gettime().
Related OS services
process management.
CPU scheduling.
Related OS services
File management.
store management
Related OS services
Device handling.
I/O management
Interrupt handling
Related OS services
system information
monitoring.
Resource management.
Scheduling
(Allocates CPU to processes).
memory allocation & storage access
manage file data in memory and disk.
Device Handling
controls I/O devices and device drivers.
memory allocation & system information
provides system status and resource details.